The Complement Rule Is Excluded Middle: Machine-Checked Probability over Intuitionistic Logic

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Abstract

Kolmogorov's axioms fix not only an arithmetic of chance but a *logic* of events: a Boolean algebra. There is no intrinsic reason for that choice. Different situations call for different logics, and quantum mechanics long ago moved probability onto non-Boolean event structures. We mechanize, in Lean 4 over mathlib, what probability becomes when the event logic is *intuitionistic*: valuations on frames (complete Heyting algebras), the logic of verifiable propositions. The development is about 3,950 lines, `sorry`-free, and its landmark theorems depend only on the three standard axioms.

Three families of machine-checked results emerge. *Hinges:* the classical complement rule $v(\neg A) = 1 - v(A)$ holds for every valuation exactly when excluded middle holds in the algebra, so all of probability's classicality is one axiom. Dynamically, Dempster's rule of conditioning agrees with Bayesian conditioning exactly when the evidence satisfies excluded middle. *Representation:* on finite frames every valuation is a classical probability mass function on points seen through an interior operator. On infinite frames this fails, Scott continuity restores it, and in general there is an unconditional atomic/diffuse decomposition. A valuation is a Dempster–Shafer belief function on its Booleanization, totally monotone, via an inclusion–exclusion identity that holds *with equality* for the intuitionistic join. *Guards:* a Sierpiński-space model built from a genuine measure shows the theory's "slack" is boundary mass, and a reduction to the halting problem shows the slack-free classical readout of a semidecidable event is uncomputable: slack is the price of a computable epistemic state. Along the way the proof assistant caught an unsatisfiable statement of the central Cox-style theorem and forced the identification of which axiom is irreducible. The analytic core (Aczél's associativity theorem, Hölder-style) is built from scratch as a dyadic-limit construction absent from mathlib.

CCS Concepts: • **Theory of computation → Proof theory**; • **Mathematics of computing → Probability and statistics**.

Keywords: intuitionistic logic, probability, Cox's theorem, belief functions, locales, Lean, mathlib, formalized mathematics

1 Introduction

Kolmogorov's axioms are usually read as fixing the arithmetic of chance: non-negativity, normalization, additivity. But they fix something else first, so quietly that it is easy to miss: a *logic*. The events form a Boolean σ-algebra, closed under complement and satisfying excluded middle, and the arithmetic is built on top of that logical substrate. The complement rule $P(\neg A) = 1 - P(A)$ is where the two layers touch. It is an arithmetical equation about a logical operation.

There is no intrinsic reason why probability should be compatible with exactly one logic. Different situations are described by different logics. Intuitionistic logic describes computation, verification, and finite-precision observation, where a proposition may be confirmable but not refutable. Linear logic describes resources, where using an assumption consumes it [Girard 1987]. Quantum logic describes measurement, where propositions fail to distribute. If probability is, as the Cox–Jaynes tradition has it, *extended logic* [Cox 1946; Jaynes 2003], the calculus of degrees of certainty over a space of propositions, then the logic of those propositions is a *parameter* of the theory, and Kolmogorov's calculus is the value of that parameter at Boolean logic.

This is not a speculative proposal, because physics has already forced the move once. Quantum probability lives on orthomodular lattices of projections rather than Boolean algebras, and Gleason's theorem [Birkhoff and von Neumann 1936; Gleason 1957] plays exactly the role there that this paper's representation theorems play here: identify the states compatible with the event logic. Nobody concludes that quantum probability is "not really probability." The conclusion is that the Boolean substrate was a modeling choice. Our question is what happens at a different, computationally motivated value of the same parameter:

> What does probability become when the logic of events is *intuitionistic*, and can the resulting theory be developed, end to end, under a proof assistant?

Intuitionistic logic is the natural first non-Boolean instance for a verification audience. Its propositions are the *verifiable* ones: semidecidable properties of programs, finite-precision observations of a real quantity, opens of a topological space. Its algebraic semantics is mature order theory, well supported by mathlib [mathlib Community 2020]: frames, that is, complete Heyting algebras, the lattices of open sets. And nothing is lost. Boolean algebras are exactly the frames where excluded middle holds, so the classical theory embeds as a special case rather than being replaced.

What changes, in one paragraph. A *valuation* on a frame Ω is a monotone, normalized, modular map $v : \Omega \to [0, \infty]$ (Section 2). Modularity is inclusion–exclusion, $v(a) +$

$v(b) = v(a \sqcup b) + v(a \sqcap b)$, and no axiom mentions complement. In a Heyting algebra the pseudo-complement satisfies $a \sqcap \neg a = \bot$ but not, in general, $a \sqcup \neg a = \top$. The complement rule therefore degrades to a derived *inequality* $v(a) + v(\neg a) \le 1$, with a gap

$$\text{slack}(a) \;=\; 1 - v(a) - v(\neg a)$$

measuring exactly the valuation of the region where excluded middle fails for a. In the concrete model where Ω is the opens of a space carrying a probability measure μ, the slack of an open U is precisely $\mu(\partial U)$, the measure of its topological boundary (Section 7): a structural quantity, not anyone's ignorance. The theory that results is recognizably the *shape* of Dempster–Shafer belief functions [Dempster 1967; Shafer 1976], but derived from a logical substrate rather than postulated, with the belief/plausibility gap given a non-epistemic reading.

What changes operationally. The statistical *operations* are logic-uniform. The conditional is still $v(a \sqcap b)/v(b)$, and the product rule and Bayes symmetry hold verbatim. What varies is the set of *identities* those operations satisfy, and with it the licensed computations. The classical case split over $\{A, \neg A\}$, and with it the usual denominator of Bayes' theorem $P(B) = P(B \mid A)\, P(A) + P(B \mid \neg A)\, P(\neg A)$, degrades to a lower bound whose deficit is a conditional slack. Marginalization is exact only over *genuine* partitions (Section 2). The complement trick $P(A) = 1 - P(\neg A)$ degrades to the interval $[\, v(A),\ 1 - v(A^c)\,]$ of width slack(A), so point statistics becomes interval statistics exactly on the events the logic leaves undecided, and collapses back to points on complemented ones. Updating itself forks. Dempster's rule and Bayesian conditioning, classically indistinguishable, give different posteriors from the same evidence unless its slack vanishes (Theorem 14). That is a modeling choice classical statistics never faces, because excluded middle collapses the fork.

...and when it does not. On finite frames every valuation *is* a classical probability mass function on points (Theorem 6). The logic changes not the underlying chance but which queries have point values: one computes $P(\Box A)$, the probability of the verifiable part. In the measure model slack is boundary mass, so a nonatomic distribution queried on intervals behaves entirely classically. The differences bite precisely at events with charged boundaries and at semidecidable hypotheses. The logic sits upstream of statistics the way a type system sits upstream of programs: the same syntax of operations, a different set of valid transformations, and some derivations that typecheck classically (the case split, the complement trick, "the" posterior) either fail or split in two.

Why a proof assistant. This subject sits at a triple point of order theory, analysis, and logic, where informal arguments have a documented history of going wrong. Cox's original theorem required repair after Halpern's counterexample [Cox 1946; Halpern 1999; Horn 2003], and the constructive setting multiplies the opportunities for smuggled classicality. Mechanization was not a formality here. It did real work. The checker refuted our first statement of the central Cox-style theorem, which was not merely unprovable but *unsatisfiable* (Section 6), and the repair identified precisely which axiom of the classical development is irreducible: the sum rule must be posited, because a disjunction's value is provably not a function of the values of its disjuncts (Theorem 15). The checker likewise refuted the folklore guess that the "classical fragment" of the theory is the ¬¬-stable propositions (it is the complemented ones, Section 2), and it forced non-vacuity guards: explicit models witnessing that every axiom set in the paper is inhabited by genuinely nonclassical instances, including one where collapse to classical logic is impossible *on pain of solving the halting problem* (Section 7).

Contributions. All results are formalized in Lean 4 over mathlib. The development is about 3,950 lines across fourteen modules, `sorry`-free, with every landmark theorem depending only on `propext`, `Classical.choice`, and `Quot.sound` (Section 8). To our knowledge it is the first mechanization, in any proof assistant, of non-additive probability over intuitionistic logic.

1. **The static hinge** (Section 3). The classical complement rule holds for *every* valuation on Ω iff excluded middle holds in Ω. The hard direction manufactures, at any failure of excluded middle, a valuation with positive slack from a prime ideal separation. All of probability's classicality is thus located in one axiom, Van Horn's negation axiom R3 in the Cox tradition [Horn 2003].
2. **Representation theorems** (Section 4). On a finite frame, every valuation *is* a classical probability mass function on the points, read through an interior operator. It is literally a mathlib `PMF`, and equivalently a finite mixture of the frame's points. On infinite frames this fails, by an explicit counterexample. Scott continuity restores it, the counterexample is itself a proved failure of Scott continuity, and in general every valuation splits unconditionally into an atomic part (always a sub-probability) and a diffuse remainder, a constructive Lebesgue-style decomposition.
3. **The Dempster–Shafer bridge, completed** (Section 5). Inclusion–exclusion is constructively innocent: it holds *with equality* for the intuitionistic join, on any frame. Read through the double-negation nucleus, it yields the full ∞-monotone tower, so a valuation restricted to its Booleanization is a bona

fide belief function. Additive probability does not become a belief function by losing an axiom, but by being observed through $\neg\neg$.

4. **The dynamic hinge** (Section 5). The two classically indistinguishable update rules of Dempster–Shafer theory, Dempster's rule of conditioning and Bayesian (geometric) conditioning, are separated constructively by a single number. For evidence b with $v(b) \neq 0$ they agree at every proposition iff $\mathrm{slack}(b) = 0$. Classicality of the *dynamics* is one instance of excluded middle, at the evidence.
5. **The Cox program, dissected** (Section 6 and Appendix A). The product-rule half of Cox's theorem is logic-free, pure analysis, and we build its analytic core, an Aczél/Hölder-style associativity representation absent from mathlib, from scratch: a normalized, strictly monotone, additive generator constructed as a dyadic limit (≈1,100 lines). The sum-rule half is logically irreducible. We exhibit a five-element frame on which a monotone normalized assignment is additive on disjoint joins yet not modular, and prove a disjunction's value is not any function of its disjuncts' values.
6. **Two grounding models** (Section 7). Gödel–McKinsey–Tarski, quantitatively: any classical probability measure, read on the opens of a space, is a valuation, and slack is boundary mass. And a computability guard: on the Sierpiński frame of one semidecidable observation, the slack-free classical readout of "machine c halts" is uncomputable by reduction to the halting problem, so any axiom forcing classical collapse demands uncomputable epistemic states.
7. **Product rigidity** (Section 10). On the chain product $\mathrm{Low}(\mathbb{N} \times \mathbb{N})$, every lower set is a finite union of rectangles, so two valuations agreeing on rectangles agree outright and the product valuation, whenever it exists, is unique with no continuity assumed. This bears directly on whether a probability *monad* exists for these valuations, a question whose hardness we delimit rather than resolve.

Section 8 reports the formalization design and the mathlib-upstreaming surface. Section 9 maps the relation to prior formalizations (ALEA, HoTT valuations, Markov categories) and to the informal literature. Section 10 returns to the opening theme, with the monad question stated precisely and the same questions posed over *linear* logic, where the event algebra is a quantale and "probability of a resource" awaits its Gleason, as the natural continuation of the program.

2 Valuations on Frames

2.1 The event logic

A *frame* (locale) is a complete lattice in which finite meets distribute over arbitrary joins. Equivalently, it is a complete Heyting algebra. Frames are the algebraic semantics of intuitionistic logic and the point-free abstraction of topological spaces. The motivating example throughout is $\Omega = \mathcal{O}(X)$, the opens of a space, with $\sqcap, \sqcup$ as intersection and union and the pseudo-complement a^{c} of an open given by the *interior* of its set complement. In mathlib this is the typeclass `Order.Frame`, and the Heyting operations come with the key asymmetry already in place: $a \sqcap a^{\mathrm{c}} = \bot$ holds in any Heyting algebra (`inf_compl_eq_bot`), while $a \sqcup a^{\mathrm{c}} = \top$ is available only in Boolean algebras. An element with $a \sqcup a^{\mathrm{c}} = \top$ is *complemented*: it satisfies its own instance of excluded middle.

2.2 The valuation structure

Definition 1 (`Valuation`). A *valuation* on a frame Ω is a function $v : \Omega \to [0, \infty]$ with $v(\bot) = 0$, $v(\top) = 1$, monotonicity, and *modularity*:

$$v(a) + v(b) \;=\; v(a \sqcup b) + v(a \sqcap b).$$

```
structure Valuation (Ω : Type*) [Order.
    Frame Ω] where
  toFun : Ω → ℝ≥0∞
  map_bot' : toFun ⊥ = 0
  map_top' : toFun ⊤ = 1
  mono'    : Monotone toFun
  modular' : ∀ a b, toFun a + toFun b
             = toFun (a ⊔ b) + toFun (a ⊓
                 b)
```

Values live in $[0, \infty]$ (`ENNReal`, extended non-negative reals) following mathlib's measure-theoretic convention, and normalization then confines them to $[0, 1]$ (`Valuation.le_one`). The definition is Kolmogorov's minus one thing: *no axiom mentions complement.* Classically that omission is invisible, because the complement rule is derivable from additivity and $a \vee \neg a = \top$. Constructively it is the whole story.

2.3 Slack, and its two independent obstructions

Modularity applied to the disjoint pair (a, a^{c}) collapses to $v(a) + v(a^{\mathrm{c}}) = v(a \sqcup a^{\mathrm{c}})$ (`add_compl_eq_sup`): the sum measures a's instance of excluded middle. Since that join need not be $\top$:

Theorem 2 (`add_compl_le_one`). *For every valuation and every a: $v(a) + v(a^{\mathrm{c}}) \leq 1$.*

This is the Dempster–Shafer sub-additivity inequality, here *derived* from the logic rather than postulated. The deficit

$$\mathrm{slack}(a) \;=\; 1 - \big(v(a) + v(a^{\mathrm{c}})\big) \;=\; 1 - v(a \sqcup a^{\mathrm{c}})$$

is the valuation of the undecided region. It decomposes canonically. The pair $(a^{\mathrm{cc}}, a^{\mathrm{c}})$, double negation against negation, is always disjoint but is a partition only under *weak* excluded middle. Running it through modularity splits the slack into two logically independent gaps:

Theorem 3 (`slack_eq_dnGap_add_deMorganGap`)**.** $\mathrm{slack}(a) = \mathrm{dnGap}(a) + \mathrm{deMorganGap}(a)$, *where* $\mathrm{dnGap}(a) = v(a^{cc}) - v(a)$ *vanishes when a is* $\neg\neg$*-stable (regular), and* $\mathrm{deMorganGap}(a) = 1 - v(a^{cc} \sqcup a^{c})$ *vanishes when weak excluded middle holds at a.*

The single Dempster–Shafer "ignorance" number is thus resolved into a regularity defect plus a De Morgan defect, each tracking a named intermediate logic.

2.4 What survives without excluded middle

Three classical pillars survive intact, and one fails in a precisely delimited way. *Disjoint additivity* is unconditional: $a \sqcap b = \bot$ implies $v(a \sqcup b) = v(a) + v(b)$. *Conditioning* is native: for $v(b) \neq 0$ the posterior $v(a \mid b) = v(a \sqcap b)/v(b)$ is again a valuation (the proof is frame distributivity), and the product rule $v(a \mid b) \cdot v(b) = v(a \sqcap b)$ and Bayes symmetry $v(a \mid b)\, v(b) = v(b \mid a)\, v(a)$ hold (`condVal_mul`, `condVal_symm`). *Total probability* holds over any genuine partition, meaning $a \sqcap a' = \bot$ and $a \sqcup a' = \top$: $v(b) = v(a \sqcap b) + v(a' \sqcap b)$. What fails is total probability over $\{a, \neg a\}$. That family does not tile $\top$, and the conditional masses fall short of $v(b)$ by a conditional slack (`cond_add_compl_le`). Bayesian updating survives. The assumption that A and $\neg A$ exhaust the world does not.

The classical fragment, a correction forced by the checker. Where does honestly additive probability live inside Ω? The natural guess is the $\neg\neg$-stable (*regular*) elements, which form a Boolean algebra by Glivenko's theorem. The guess is *wrong*, and the failed proof said why: the valuation's modularity governs the frame join $\sqcup$, while the regular elements' Boolean join is the regularized $(\cdot \sqcup \cdot)^{cc}$. A regular but uncomplemented element (the open ray $(-\infty, 0)$ in the opens of $\mathbb{R}$, under an atomic measure at 0) still carries slack. The correct classical fragment is the *complemented* elements, where $v(a) + v(a^{c}) = 1$ holds on the nose (`add_compl_eq_one_of_complemented`). On a connected locale that fragment can be as small as $\{\bot, \top\}$. In the Boolean limit every element is complemented, the slack vanishes identically, and Kolmogorov is recovered (`classical_additivity`, `classical_slack_zero`).

Mixtures. Valuations are closed under finite convex combination (`Valuation.mix`), with every axiom inherited termwise, because modularity is linear. This innocuous fact becomes a characterization in Section 4.

3 Where Classicality Lives: the R3 Hinge

Van Horn's repaired Cox theorem [Horn 2003] axiomatizes plausibility over classical propositions. Its negation axiom, called *R3*, posits that the plausibility of $\neg A$ is a fixed nonincreasing function S of the plausibility of A. Consistency forces $S \circ S = \mathrm{id}$, an involution, and thence the complement rule $p \mapsto 1 - p$. An involutive negation *is* double-negation elimination, so R3 is where classical logic enters every known Cox-style derivation. We isolate the exact content of R3's conclusion as a predicate on valuations and prove it equivalent to excluded middle.

```
def HasClassicalNegation (v : Valuation Ω)
    : Prop :=
  ∀ a, v a + v aᶜ = 1
```

Theorem 4 (the R3 hinge)**.** *For a frame Ω, every valuation on Ω has classical negation* iff $a \sqcup a^{c} = \top$ *for every* $a \in \Omega$.

The forward direction (`hasClassicalNegation_of_em`) is a calculation. The content is the converse (`em_of_forall_hasClassica`
wherever excluded middle fails, a valuation with genuine slack must be *manufactured*. Suppose $a \sqcup a^{c} \neq \top$. The prime ideal separation theorem (mathlib's `DistribLattice.prime_ideal_of_disjoint_filter_ideal`) yields a prime ideal J containing $a \sqcup a^{c}$ but not $\top$. The indicator of J's complement,

$$v(x) = \begin{cases} 0 & x \in J \\ 1 & x \notin J, \end{cases}$$

is a valuation. Modularity in the only nontrivial case $(x, y \notin J)$ is *exactly* primeness, $x \sqcap y \in J \Rightarrow x \in J \vee y \in J$. And this valuation assigns $v(a) + v(a^{c}) = 0 \neq 1$. So the complement rule is not one convention among many. Quantified over all valuations, it is a statement *about the algebra*, and the statement is excluded middle.

Sharp valuations are the points of the locale. The construction above is not ad hoc. It is one half of an exact correspondence. Call v *sharp* if it is $\{0, 1\}$-valued, a state of full certainty.

Theorem 5 (`sharp_iff_point`)**.** *v is sharp iff v is the complement-indicator of a prime ideal. The maps are mutually inverse: the ideal recovered from v is its zero-set, prime exactly because v is sharp.*

In locale-theoretic terms, the certainty limit of the probability theory recovers the *points* of the locale. (These are the finitely prime points. The completely prime points would correspond to sharp valuations that are also Scott-continuous, a refinement we note but do not formalize.) The hinge and the correspondence together are the paper's static answer to "where does classicality live": in one axiom, whose semantic shadow is point-hood.

4 Representation: Points, Masses, Mixtures

Is intuitionistic probability a new animal, or classical probability seen at an angle? The answer is exact: on finite frames the latter, in general the former, with Scott continuity marking the boundary.

4.1 Finite frames: every valuation is a PMF on points

Every finite frame is the lattice of *lower sets* of a finite poset P, by Birkhoff duality. We work with `LowerSet P` directly and cite the duality rather than mechanize it. Define the *mass* of a point as the discrete derivative of v:

```
def Valuation.mass (v : Valuation (
    LowerSet P)) (p : P) :
    ℝ≥0∞ := v (LowerSet.Iic p) - v (
        LowerSet.Iio p)
```

Theorem 6 (finite representation, `eq_sum_mass`, `sum_mass`). *For finite P and any valuation v on $\mathrm{Low}(P)$: $v(U) = \sum_{p\in U} \mathrm{mass}(p)$ and $\sum_p \mathrm{mass}(p) = 1$.*

The proof peels a maximal point p of U at a time. Modularity applied to $(U \setminus \uparrow p, \downarrow p)$ shows the increment is exactly $\mathrm{mass}(p)$, and a strong induction on $|U|$ finishes. The arithmetic is in $[0, \infty]$ with truncated subtraction, where the bookkeeping is real work. Two packagings make the classicality literal. `Valuation.toPMF` exhibits mass as a mathlib `PMF` on P. And since lower sets are the opens of the Alexandrov topology, where interior is the identity, the theorem reads: every valuation on a finite frame is a classical probability seen through the interior operator, $v = \mu \circ \Box$. This is the converse, in the finite case, of the measure-theoretic bridge of Section 7.

Dually, with δ_p the sharp *point-valuation* $\delta_p(U) = [p \in U]$:

Theorem 7 (mixture characterization, `eq_mix_deltaPoint`). *On a finite frame, $v = \sum_p \mathrm{mass}(p) \cdot \delta_p$: valuations are exactly the finite convex mixtures of point-valuations.*

Modularity is thereby *characterized* as closure under mixing certainties. This is the constructive answer to what the sum rule axiomatizes, given that it cannot be derived (Section 6).

4.2 The infinite boundary: enter Scott continuity

Both halves of the finite theorem fail at the first infinite frame, and the failure is informative.

Theorem 8 (obstruction, `exists_valuation_not_point_representable`). *On $\mathrm{Low}(\mathbb{N})$, the indicator of $\top$ is a valuation with $\mathrm{mass}(p) = 0$ at every point yet $v(\top) = 1$. The unit of mass escapes to the missing point at infinity.*

Theorem 9 (recovery, `eq_tsum_mass_of_scott`, generalized by `isPurelyAtomic_of_scott`). *On $\mathrm{Low}(\mathbb{N})$, Scott continuity restores point-representability: if $v(\top) = \bigsqcup_n v(\mathrm{Iic}\, n)$ then $v(\top) = \sum'_n \mathrm{mass}(n)$. The counterexample of Theorem 8 is itself a proved failure of Scott continuity (`topIndicator_not_scott`). More generally, on any poset all of whose principal downsets are finite, Scott continuity implies zero diffuse part.*

Theorem 10 (general decomposition, `tsum_mass_le`). *On $\mathrm{Low}(P)$ for arbitrary P, with no finiteness and no continuity assumption: $\sum'_p \mathrm{mass}(p) \le v(\top)$.*

Every valuation thus splits unconditionally into an *atomic* part carried by points, always a sub-probability, and a *diffuse* remainder $v(\top) - \sum'_p \mathrm{mass}(p) \ge 0$. This is a constructive Lebesgue-style decomposition. The finite and Scott cases are the purely atomic extreme, and the $\top$-indicator is the purely diffuse one. The proof of Theorem 10 re-runs the maximal-point peel over arbitrary finite subsets, bounded by the whole, and passes to the supremum. It uses neither excluded middle nor choice beyond mathlib's baseline. Representing the diffuse part on non-spatial frames, where, tellingly, the obstruction is *too few points*, the same phenomenon as undecidability in Section 7, is the open frontier (Section 10).

5 The Dempster–Shafer Reading, Completed

Sections 2–4 keep meeting the *shape* of Dempster–Shafer (DS) theory [Dempster 1967; Shafer 1976]: sub-additive complements, a belief/plausibility interval, mass functions. This section makes the identification exact in both statics and dynamics, and locates, each time, the precise point where excluded middle is the difference.

5.1 Statics: total monotonicity via constructive inclusion–exclusion

The $\neg\neg$-stable (*regular*) elements of a frame form a Boolean algebra with meet $\sqcap$ and join $(a \sqcup b)^{cc}$ (Glivenko). Call it the *Booleanization*. A DS belief function is a capacity on a Boolean algebra that is n-monotone for every n: $\mathrm{Bel}(\bigvee_i A_i) \ge \sum_{\emptyset\ne T} (-1)^{|T|+1} \mathrm{Bel}(\bigwedge_{i\in T} A_i)$. The 2-monotone case is a one-liner from modularity (`two_monotone`). The full tower follows from a fact that is interesting in its own right:

Theorem 11 (inclusion–exclusion, `inclusion_exclusion`). *For every valuation on every frame and every finite family $a : \iota \to \Omega$, inclusion–exclusion holds* with equality *for the frame join:*

$$v\Big(\bigsqcup_{i\in s} a_i\Big) = \sum_{\emptyset\ne T\subseteq s} (-1)^{|T|+1}\, v\Big(\bigsqcap_{i\in T} a_i\Big).$$

(Stated in Lean additively, with the odd-cardinality half of the signed sum set against the even-cardinality half, so that no $[0, \infty]$ subtraction occurs. The proof is induction with modularity and frame distributivity.)

Theorem 12 (total monotonicity, `infty_monotone`). *For every valuation, finite family, and n:*

$$\sum_{\emptyset\ne T} (-1)^{|T|+1} v\Big(\bigsqcap_T a_i\Big) \le v\Big(\big(\bigsqcup_i a_i\big)^{cc}\Big).$$

Read on the Booleanization, where $(\cdot)^{cc} \circ \sqcup$ is the join and meets of regulars are regular, this is verbatim the ∞-monotone *tower: v restricted to its Booleanization is a bona fide belief function.*

The proof of Theorem 12 from Theorem 11 is one line: monotonicity along $a \le a^{cc}$. The conceptual point runs opposite to the usual telling. Inclusion–exclusion does *not* fail constructively. It is exact for the intuitionistic disjunction, and total monotonicity is what that equality looks like through the double-negation nucleus: rounding a join up to its regular hull can only add value, and the defect is exactly the double-negation gap of Section 2. Additive probability does not *become* a belief function by losing an axiom. It becomes one by being *read through* $\neg\neg$. With $\mathrm{Pl}(a) = 1 - v(a^c)$, the belief–plausibility interval has width exactly the slack (`plausibility_sub_self`). The DS "ignorance" is boundary measure, not psychology.

5.2 Dynamics: Dempster against Bayes, separated by one number

DS theory carries two update rules that coincide classically. *Geometric* conditioning renormalizes belief, and under the DS dictionary it *is* the localic posterior $v(a \sqcap b)/v(b)$ of Section 2. *Dempster's rule of conditioning* [Shafer 1976] discards the mass incompatible with the evidence and renormalizes by its plausibility:

$$v(a \,\|\, b) \;=\; \frac{v(a \sqcup b^c) - v(b^c)}{1 - v(b^c)}.$$

Both are again valuations (`dempsterCond`, whose modularity is the frame distributivity $(a \sqcup b^c) \sqcap (a' \sqcup b^c) = (a \sqcap a') \sqcup b^c$). Their disagreement is governed by a single structural quantity, the *excluded-middle gap* $\mathrm{emGap}(a, b) = v(a) - v(a \sqcap (b \sqcup b^c))$, the mass of a stranded outside b's instance of excluded middle:

Lemma 13 (`sup_compl_decomp`). $v(a \sqcup b^c) = v(b^c) + v(a \sqcap b) + \mathrm{emGap}(a, b)$. *Dempster's numerator exceeds Bayes's by exactly the gap.*

Theorem 14 (the conditioning hinge, `dempsterCond_eq_condVal_iff_slack`). *For $v(b) \neq 0$: Dempster conditioning on b agrees with Bayesian/geometric conditioning on b at* every *proposition iff* $\mathrm{slack}(b) = 0$.

Zero slack at the evidence kills the gap for every a and makes the two normalizers coincide, since $1 - v(b^c) = v(b)$. Conversely, testing at $a := b$, geometric conditioning gives certainty 1 while Dempster gives $v(b)/(1 - v(b^c))$, which is 1 only if the slack vanishes. Theorem 14 is the dynamic companion of Theorem 4. The static hinge locates classicality in the complement rule, and the dynamic hinge locates it in the agreement of the two DS update rules: one instance of excluded middle, at the evidence. In the measure model, Dempster and Bayes disagree exactly when the evidence has a charged boundary.

6 The Cox Program, Dissected

Cox's theorem [Cox 1946; Jaynes 2003] derives the probability calculus from desiderata on plausibility. After Halpern's counterexample [Halpern 1999], its repaired forms [Arnborg and Sjödin 2000; Horn 2003] fix the calculus up to monotone regraduation. Every known derivation is classical, and Section 3 showed why: the negation axiom R3 *is* excluded middle. What survives of the program when R3 is dropped? Mechanization forced a sharper answer than we had conjectured, in three parts. The product rule is logic-free. The sum rule is irreducible. And the naive constructive statement is unsatisfiable.

6.1 A Cox structure without negation

```
structure CoxModel (Ω : Type*) [Order.
    Frame Ω] where
  pl : Ω → Ω → ℝ              -- pl a b :
      plausibility of a given b
  mono_left : ∀ c, Monotone fun a => pl a
      c
  pl_bot : ∀ c, pl ⊥ c = 0
  pl_top : ∀ c, pl ⊤ c = 1
  F : ℝ → ℝ → ℝ               -- conjunction
       functional
  conj : ∀ a b c, pl (a ⊓ b) c = F (pl a (
      b ⊓ c)) (pl b c)
  F_assoc : ∀ x y z, F (F x y) z = F x (F
      y z)
  F_strictMono_left : ∀ y, 0 < y →
      StrictMono fun x => F x y
```

The deliberate omission is any field relating $\mathrm{pl}(a^c, b)$ to $\mathrm{pl}(a, b)$. That missing field is exactly R3. Two details are themselves checker-forced. Strict monotonicity of F must be restricted to $0 < y$, because the boundary axioms force $F(x, 0) = 0$ for all x, so the unrestricted axiom makes the structure *uninhabited* and every theorem about it vacuous. We guard against this permanently: `coxModelENNReal` is an explicit model on the non-Boolean chain $[0, \infty]$, built without classical logic (deciding $a = \top$ there is a constructor check), so the axiom class is inhabited by a genuinely non-classical instance.

6.2 The sum rule is irreducible

Classically, additivity is *derived*: R3 turns disjunction into negated conjunction via De Morgan, and the product rule does the rest. Constructively that route is closed, since $a \vee b \neq \neg(\neg a \wedge \neg b)$ in general. And this is not an artifact of one proof strategy:

Theorem 15 (`modularity_irreducible`, `no_disjunction_functional`
There is a five-element frame (the lower sets of the poset $o < a$, $o < b$ with a, b incomparable) and a monotone, normalized assignment q on it that is additive on disjoint

joins yet not modular. Moreover no binary operation S on values satisfies $q(x \sqcup y) = S(q(x), q(y))$ for all x, y. The pairs $(\downarrow a, \downarrow a)$ and $(\downarrow a, \downarrow b)$ have identical value-pairs $(\frac{1}{2}, \frac{1}{2})$ but joins valued $\frac{1}{2}$ and 1.

A disjunction's plausibility is not a function of its disjuncts' plausibilities, unlike a conjunction's, which the `conj` field reduces to values via conditioning. So a constructive Cox theorem *must* posit the sum-rule half. Modularity is the posit, and Theorem 7 is its independent justification: modularity says exactly that the calculus is closed under mixing certainties.

6.3 The theorem, corrected and proved

Our original target statement asked for a strictly monotone regraduation $g : \mathbb{R} \to [0, \infty]$ with $g(0) = 0$. The checker refused, for a reason no referee had caught on paper: strict monotonicity forces $g(-1) < g(0) = 0$, impossible in $[0, \infty]$. The statement was not too hard. It was *unsatisfiable*, and the repair (strictness only on $[0, 1]$, where plausibilities live) is exactly the domain hygiene that Halpern's critique of the classical theorem turns on.

Theorem 16 (`constructive_cox`)**.** *Every Cox model whose unconditional plausibility is modular regraduates to a valuation: there exist v : Valuation Ω and g strictly monotone on $[0, 1]$ with $g(0) = 0$, $g(1) = 1$, and $v(a) = g(\mathrm{pl}(a, \top))$ for all a.*

The regraduation for the sum-rule half is undemanding (g = `ENNReal.ofReal`). The substance lives in the product-rule half, showing F regraduates to multiplication, which is Aczél's associativity theorem [Aczél 1966], a statement of pure analysis whose type never mentions Ω. The constructive/classical divide is therefore *entirely* in the sum rule. mathlib has no Aczél theorem, no Hölder-style embedding for ordered semigroups on intervals, and no Cauchy functional equation under monotonicity (only under continuity), so we build the analytic core from scratch (≈1,100 lines): a bundled `Scale` of Aczél hypotheses, a generator defined as the dyadic limit $g(b) = \bigsqcup_n (\text{count of } 2^n\text{-th roots under } b)/2^n$, and the Hölder-style conclusion, a normalized, strictly monotone additive generator and hence a strictly monotone multiplicative one, on the positive cone (`exists_mul_generator`, `aczelStatement_cone`). The Archimedean property is *derived* from continuity and positivity rather than assumed, and the interval form of divisibility is likewise a consequence of the intermediate value theorem (`exists_diag_eq`), though the bundled `Scale` takes divisibility as a hypothesis. Appendix A gives the construction. The residual analytic gap, continuity of the generator and extension below the unit, is stated honestly there and in Section 10, and is witnessed non-vacuously by the log-sum-exp archetype, for which the global continuous generator is exp.

7 Two Grounding Models

Two concrete models anchor the abstract theory. One shows every classical measure *contains* an intuitionistic probability. The other shows the intuitionistic theory cannot collapse back without solving the halting problem.

7.1 Measures through the interior operator

Gödel–McKinsey–Tarski interpret intuitionistic logic in modal S4, semantically: propositions are *open* sets, and $\Box$ is topological interior [Gödel 1933; McKinsey and Tarski 1944]. Our valuations are the quantitative face of that translation:

Theorem 17 (`Measure.toValuationOpens`)**.** *Let μ be a probability measure on a space X. Then $U \mapsto \mu(U)$ on the frame $\mathcal{O}(X)$ is a valuation. Equivalently, $v = \mu \circ \mathrm{int}$ as a function of arbitrary sets, that is, $v(A) = P(\Box A)$.*

Modularity is $\mu(U \cup V) + \mu(U \cap V) = \mu(U) + \mu(V)$. The slack of an open U is then $1 - \mu(U) - \mu(\mathrm{int}\, U^{\complement})$, which is the measure of the boundary ∂U. *Slack is boundary mass.* (The general boundary identity is elementary measure theory. The development proves the complement sub-additivity inequality in general, `interiorMeasure_add_compl_le`, and mechanizes the boundary identity end to end in the Sierpiński model below.) The DS "ignorance" of Section 5 is, in this model, the measure charged to the frontier where a finite-precision observer can never decide membership, a structural quantity with no epistemic subject. This also exhibits the modal identification Bel $= P(\Box \cdot)$ of the DS literature [Ruspini 1986] arising from the logic itself. Theorem 6 is the converse in the finite case, and the general converse is the open frontier of Section 4.

7.2 The computability guard

The design constraint on every axiom set in this paper is that it must not silently force slack $\equiv 0$. Otherwise the "constructive" theory is Kolmogorov in disguise. Our guard is a model in which collapse is not merely false but *uncomputable*.

A semidecidable proposition is an open in the Sierpiński topology on `Prop`: verifiable by a finite computation, never refutable by one. On the three-open frame $\mathcal{O}(\mathbb{S}) = \{\bot, \mathit{halts}, \top\}$ we prove the Σ_1 asymmetry as a theorem, $\mathit{halts}^{\mathrm{c}} = \bot$ (every nonempty open of $\mathbb{S}$ contains True, so nothing partially refutes a semidecidable event), and build the valuation from a genuine measure $p \cdot \delta_{\mathsf{True}} + (1 - p) \cdot \delta_{\mathsf{False}}$ through the bridge above. The slack computes end to end (`slack_eq_boundary`):

$$\mathrm{slack}(\mathit{halts}) = 1 - p = \mu\big(\partial\, \{q \mid q\}\big) = \mu\{\mathsf{False}\},$$

the mass of the point where the computation stays silent forever.

The guard itself is grounded in mathlib's computability theory. For a code c and input n, the *sharp classical readout* assigns the slack-free valuation answering "does c halt on

n?" with certainty, weight 1 or 0 by the actual halting fact. This family is classically well defined, and:

Theorem 18 (`sharpReadout_not_computable`)**.** *Deciding whether the sharp classical readout assigns belief* 1 *to the halting event is exactly the halting problem, hence not computable (*`ComputablePred.halting_problem`*).*

Any axiom strong enough to force classical collapse therefore demands of its models an uncomputable epistemic state. Slack is not a defect the constructive theory should apologize for. It is the price of a computable one. This guard plays for the sum rule the role the explicit `CoxModel` instance plays for the product rule (Section 6): each axiom set in the paper ships with a machine-checked witness that it has non-degenerate models.

8 Formalization Design and Reuse

Shape of the development. Fourteen Lean 4 modules, about 3,950 lines, over a pinned `mathlib` (both at v4.31.0): the core valuation theory and GMT bridge, the DS layer (2-monotonicity, inclusion–exclusion and the ∞-monotone tower, the two conditionings), the representation suite (finite, obstruction, Scott, general decomposition), the Cox layer (`CoxModel`, sum-irreducibility, the corrected theorem), the Aczél development, the two guards (halting chain, Sierpiński model), the product rigidity theorem on chain products, and the countable-mixture composition laws. The build is clean, with no `sorry` and no warnings, and every theorem named in this paper reports exactly `[propext, Classical.choice, Quot.sound]` under `#print axioms`.

Meta-logic against object logic. Lean's logic is classical, while our *object* of study is the frame. This division of labor is deliberate and, we found, the pragmatic sweet spot. Constructivity claims attach to the algebra (no theorem assumes $a \sqcup a^{c} = \top$), while the meta-theory may reason classically *about* it. The counterexample valuations of Sections 4 and 6 use classical case splits legitimately, since a counterexample is a classical fact about the theory. Where the distinction has content we track it: the non-vacuity witness for `CoxModel` was re-based from `Prop` (which imports excluded middle through the back door) to the chain $[0, \infty]$, where the needed case split is a constructor check.

What the checker contributed. Beyond the usual assurance, five findings in this paper originate in failed proofs. The naive `constructive_cox` statement was unsatisfiable (Section 6). Unrestricted strict monotonicity for the conjunction functional was vacuous (Section 6). The regular-elements guess for the classical fragment was false (Section 2). Our first Aczél hypothesis bundle omitted order-preservation, which the additivity proof needs (Appendix A). And our expected counterexample to product uniqueness was refuted by the rigidity that became Section 10's theorem. We regard these as the paper's best evidence that this subject specifically repays mechanization.

mathlib surface. The development consumes order theory (`Order.Frame`, Heyting algebras, prime ideal separation), `ENNReal` arithmetic, measure theory for the bridge, `PMF`, and the computability library (`Nat.Partrec.Code`, `ComputablePred`). Candidates for upstreaming, in decreasing generality: the Aczél/Hölder generator construction of Appendix A (mathlib has no associativity-representation theorem, no monotone Cauchy functional equation, and no t-norm theory), valuations on frames with the mass/representation API, and the inclusion–exclusion identity for modular functions on distributive lattices (Theorem 11), which is independent of probability. $[0, \infty]$'s truncated subtraction deserves a note. It forces additive normal forms, with every identity stated so that no subtraction occurs. This reads as a constraint but functioned as a correctness discipline: several informal manipulations of the DS literature do not survive it unchanged.

Artifact. The artifact (anonymized supplement) builds with a two-command recipe on the pinned toolchain and re-checks every theorem, including the axiom audit.

9 Related Work

Informal precedents. That probability might be paired with intuitionistic logic is an old if scattered thought. Weatherson develops intuitionistic probability axiomatically and defends it with Dutch-book arguments [Weatherson 2003]. Paris treats non-classical uncertain reasoning broadly [Paris 1994]. The identification Bel $= P(\Box\,\cdot)$ connecting belief functions to modal logic is classical [Dempster 1967; Ruspini 1986; Shafer 1976]. On the localic side, valuations on frames are standard objects of constructive measure theory, with Vickers's work and the Coquand–Spitters development as canonical references [Coquand and Spitters 2009; Vickers 2008], normally with Scott continuity built in and with additive, Riesz-flavored goals. What we take from Section 4 is that Scott continuity is a substantive assumption (it decides point-representability on the frames we examine), so our base structure omits it. The Cox line, through Cox, Jaynes, Halpern's counterexample, and the Van Horn and Arnborg–Sjödin repairs [Arnborg and Sjödin 2000; Cox 1946; Halpern 1999; Horn 2003; Jaynes 2003], is classical throughout. Van Horn himself observes that dropping R3 yields a two-number theory, which is exactly the belief/plausibility pair our slack separates. To our knowledge, no prior work, informal or formal, states either hinge theorem: complement rule $\Longleftrightarrow$ excluded middle (Theorem 4), or Dempster = Bayes $\Longleftrightarrow$ zero slack at the evidence (Theorem 14).

Prior formalizations. Three bodies of mechanized work are adjacent, and none overlaps. ALEA [Audebaud and Paulin-Mohring 2009] formalizes a discrete distribution monad in Coq for randomized programs: additive probability, no event logic in our sense. Recent work in cubical Agda formalizes synthetic probability via Markov categories [Sargsyan et al. 2023], where again the objects are additive distributions. Closest is the HoTT/Coq line of Faissole and Spitters on *valuations* over lower reals [Faissole and Spitters 2017], done constructively with Scott continuity and aimed at integration (Riesz) and a Giry-style valuation *monad*. Their valuations are the additive, continuous cousins of ours, and their monad is the Scott-continuous instance of the question examined in Section 10. The delta of this paper is not the word "valuation" but the development on top of it: the non-additive complement regime, slack and its decomposition, both hinge theorems, the DS bridge with total monotonicity and the two conditionings, the representation/mixture/Scott-atomic suite, the corrected Cox theorem with its irreducibility counterexamples, the Aczél construction, product rigidity, and the computability guard. We claim first mechanization for that development, not for valuations on locales as such.

Non-Boolean probability elsewhere. Quantum probability on orthomodular lattices, with Gleason's theorem [Birkhoff and von Neumann 1936; Gleason 1957], is the long-standing precedent for re-basing probability on a non-Boolean event logic. Free probability, fuzzy measures, and imprecise probability [Walley 1991] vary other parameters (independence, additivity, single-valuedness). Our variation is specifically the *logic*, holding the Cox-style methodology fixed. Within DS theory, the two conditionings and their inequivalence are classical [Fagin and Halpern 1991; Shafer 1976]. What is new in Theorem 14 is the exact logical characterization of their agreement, and its mechanization.

10 Conclusion: Probability with the Logic as a Parameter

We set out from the observation that Kolmogorov's axioms fix a logic before they fix an arithmetic, and that nothing in the concept of graded belief privileges the Boolean choice. The machine-checked results bear the framing out with a precision informal treatments have not reached. The entire classical surplus is one axiom, whose statical face is the complement rule (Theorem 4) and whose dynamical face is the agreement of Dempster and Bayesian conditioning (Theorem 14). Everything else survives the intuitionistic substrate intact: conditioning, Bayes symmetry, disjoint additivity, total probability over genuine partitions, and inclusion–exclusion itself (Theorem 11). What the weaker logic adds is structure, not damage: slack with its two-gap decomposition, the atomic/diffuse divide, the belief-function reading through $\neg\neg$, and a computability-theoretic reason the additions are not optional.

Open problems. Three are mathematical. First, representing the diffuse part on non-spatial frames (the localic Riesz direction). Second, closing the analytic residue of the Aczél development, namely continuity of the generator and the off-cone extension (the cone results and the log-sum-exp archetype are done). Third, a genuine uniqueness theorem: deriving modularity from a more primitive desideratum, knowing (Theorem 15) that it cannot come from disjunction data alone, with the mixture characterization (Theorem 7) as the current best justification. A fourth is foundational hygiene at scale: a machine-checked Dutch-book theorem in the style of [Weatherson 2003], which to our knowledge no proof assistant has for *any* logic.

The functor is literal, the monad is open. Within a fixed doctrine the connection to the categorical picture is concrete. Pushforward along frame homomorphisms preserves all four valuation axioms, making $\mathrm{Val}(-)$ a functor on locales, the localic face of the functor part of the Giry monad [Giry 1982]. This is a routine verification that we state but have not mechanized. The point valuations δ_p of Section 4 are the evident unit (these *are* formalized), and Theorem 6 shows that on finite frames Val agrees, on objects, with the classical finite-distribution monad on points. The *multiplication* is where the substance is. Flattening a valuation of valuations is integration, hence product valuations and Fubini, and by Kock's identification of commutative monads with Fubini theorems [Kock 2012], whether statistics-as-kernel-composition transports to a weaker logic is precisely the question whether that logic's valuation functor is a commutative monad. One machine-checked result of this paper bears on it directly, and it surprised us: on the chain product, the question of *which* product valuation to take does not arise. The frame $\mathrm{Low}(\mathbb{N}\times\mathbb{N})$ is *rigid*. Every lower set is a *finite* union of rectangles, because column heights are antitone and an antitone map out of $\mathbb{N}$ is a finite step function (`exists_rect_decomposition`). By the inclusion–exclusion equality (Theorem 11), any two valuations agreeing on rectangles therefore agree outright (`Valuation.product_unique`): the product valuation, whenever it exists, is unique, with no continuity assumed. We had expected the opposite, diffuse mass free to choose among the infinities of the product, and our candidate counterexamples all died on the same rock: lower sets of chain products detect only finitely much of the boundary. Whether non-uniqueness, that is, genuine freedom in the multiplication, occurs on frames that are not finitely rectangle-generated is open. Fubini for non-additive integration is where the trouble is classically located [Denneberg 1994; Ghirardato 1997].

The hard problem. The question *upstream* of the multiplication deserves to have its hardness named rather than implied, and what follows is our assessment rather than a theorem. A monad needs a category for Val to be an endofunctor on. The space of valuations must itself be an object, a "space" whose opens one can speak of. For Scott-continuous valuations that space is a locale and monad structure is available, constructively, through the probabilistic powerdomain line [Coquand and Spitters 2009; Jones and Plotkin 1989; Vickers 2008]. Yet *even there*, exhibiting a cartesian closed category closed under the construction is the Jung–Tix problem, open since 1998 [Jung and Tix 1998]. Dropping continuity makes the difficulty prior rather than merely worse: the standard presentations of the valuation space use continuity essentially. One route to a well-posed question does exist. The localic treatments present the space of continuous valuations as the classifying locale of a propositional geometric theory [Coquand and Spitters 2009; Vickers 1989], with generators asserting $q < v(a)$ for rational q and axioms for monotonicity, normalization, modularity (expressible with rational witnesses and countable disjunctions), and continuity. Deleting the continuity axiom leaves a geometric theory, so a classifying locale still exists on general grounds, and classically its points are exactly the valuations of this paper. We have verified none of the structure this candidate would need. The unit looks unproblematic, but the multiplication requires integrating the evaluation maps against a merely modular, discontinuous valuation, and there the sum of two lower-real-valued maps is an infinite join, exactly where the diffuse part of Theorem 8 leaks. We offer the formulation as a target, not a result.

The strategy with the best modern track record is to carve a fragment rather than to conquer the whole. On the measure-theoretic side, the failure of cartesian closure for measurable spaces was answered by quasi-Borel spaces [Heunen et al. 2017]. On the domain-theoretic side, commutativity of the full valuations monad on DCPO is open, and the response is a hierarchy of commutative *submonads* generated from simple valuations [Jia et al. 2021]. Our decomposition theorem names the corresponding fragment here: the atomic part. As a first machine-checked step in that direction we prove that the countably-presented fragment composes with no space structure at all. A countable mixture of valuations is a valuation (`mixCountable`), a mixture of mixtures flattens to a mixture with product weights, which is discrete Chapman–Kolmogorov (`mixCountable_mixCountable`), and the one-point mixture is the identity (`mixCountable_unique`), all by the unconditional Fubini of $[0, \infty]$ sums over sigma types. This is associative, unital kernel composition for countably-atomic data. It is not yet a monad: the presentations are not canonical, we do not quotient by "presents the same valuation," and the diffuse part is exactly what escapes every countable presentation. We conjecture the ledger will stay lopsided for some time. The unit, the finite level, and countable kernel composition are proved. Product uniqueness on rigid frames is proved. The general multiplication is open. And the ambient category is a formulation problem before it is anything else.

Across doctrines, by contrast, we make no functoriality claim. There is no ambient category in which "logic $\mapsto$ its probability theory" is a functor, and each column requires rederiving what a state *is* from the desiderata. What is uniform across columns is the *method*, and this paper's execution of it is the Heyting column.

The program. If the logic is a parameter, this paper computed one new value of it. The table wants filling in. Linear logic is the most inviting column for this community. Its event algebras are quantales, its conjunction is resource-sensitive, and a "valuation" there should grade the *consumption* of evidence. The analogue of Gleason's theorem or of our representation theorems (which states are compatible with the algebra?) is wide open, as is the right analogue of slack for the failure of contraction. The probabilistic semantics of linear logic that already exist, probabilistic coherence spaces [Danos and Ehrhard 2011], start from the additive side. A Cox-style derivation over a quantale would start from the logic. The methodology of this paper, to state the classical desiderata, delete the axioms the logic no longer licenses, and let a proof assistant referee what survives, what splits, and what becomes unsatisfiable, is, we suggest, the right instrument for that program. Each cell of the table is exactly the kind of delicate, counterexample-rich boundary work that mechanization is for.

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A The Aczél Development

This appendix sketches the from-scratch construction behind the product-rule half of Theorem 16. The body stands without it. The target shape is Aczél's associativity theorem [Aczél 1966]: a suitably regular associative $F : \mathbb{R} \to \mathbb{R} \to \mathbb{R}$ regraduates to multiplication, $G(F\,x\,y) = G(x) \cdot G(y)$ for a strictly monotone G. mathlib offers no route. It has no associativity representation, no Hölder-style embedding of ordered semigroups on intervals, and the Cauchy functional equation only under continuity, not monotonicity.

Hypotheses. A `Scale` bundles: associativity of F, *divisibility* $\forall c\, \exists x,\ F\,x\,x = c$, positivity on the cone $x < F\,x\,c$, joint order-preservation, joint continuity, a unit u, and identity at $-\infty$, meaning $F\,x\,z \to x$ as $z \to -\infty$. Order-preservation was absent from our first bundle, and the additivity proof needs it, a definition-check finding (Section 8). Divisibility is taken as a hypothesis in the bundle, although its interval form is a consequence of the intermediate value theorem (`exists_diag_eq`). Non-vacuity: *log-sum-exp*, $F\,x\,y = \log(e^x + e^y)$ with $u = 0$, satisfies every field (`logSumExpScale`), and for it the intended generator is globally exp.

Construction. Write $a^{(n)}$ for n right-associated copies of a under F (`Fpow`). Associativity alone makes $n \mapsto a^{(n)}$ a semigroup morphism from $(\mathbb{N}, +)$, with no commutativity needed. Commutativity is a *corollary* of the generator, not a prerequisite. One fact usually assumed is here *derived*: unboundedness of the iterates. Were they bounded, monotone convergence would give a fixed point $F\,L\,a = L$, contradicting positivity, and unboundedness yields the Archimedean property. Divisibility supplies a tower of 2^n-th roots $u \succ u^{1/2} \succ u^{1/4} \succ \cdots$ (`droot`) with the coherence law that 2^ℓ fine copies recompose one coarse mark. Counting how many copies of the n-th root fit under b (`dcount`, well defined by the Archimedean property) gives dyadic approximants $g_n(b) = \mathrm{dcount}(b, n)/2^n$. The refinement sandwich $2N + 1 \le N' \le 2N + 2$ makes them monotone and Cauchy, and the generator is their limit, $g(b) = \bigsqcup_n g_n(b)$.

Results (all `sorry`-free, standard axioms). On the cone $\{x \mid u \le x\}$ we prove three things. Normalization: $g(u) = 1$. Additivity: $g(F\,x\,y) = g(x) + g(y)$, by squeezing super- and sub-additive counting bounds ($\mathrm{dcount}\,x + \mathrm{dcount}\,y + 1 \le \mathrm{dcount}(F\,x\,y) \le \mathrm{dcount}\,x + \mathrm{dcount}\,y + 3$) against a vanishing 2^{-n} slack. And *strict* monotonicity, proved without any continuity of g by a count-amplification argument: a shrinking root inserted between $x < y$ contributes a full extra block of 2^{n-M} copies. Exponentiating gives the multiplicative form, a strictly monotone G with $G(F\,x\,y) = G(x)\,G(y)$ on the cone (`exists_mul_generator`), which is the conclusion shape of Aczél's theorem restricted to the cone (`aczelStatement_cone`). A bounded reorientation $\bar{G} = \exp(-g)$ transports it to the $[0, 1]$, conjunctive orientation of Section 6.

Honest residue. The constructed g is discontinuous at the unit (it is 0 below u), so matching the classical statement verbatim still needs the off-cone extension (a group completion) and continuity of the extension. That is genuine analysis, open here, but *witnessed*: for the log-sum-exp archetype the global continuous generator exists (`hasOrderedGenerator_logSumExp`). Of Hölder's classical hypotheses, the Archimedean property is a theorem in this development rather than an assumption, and the interval form of divisibility follows from the intermediate value theorem. Only the embedding's global regularity remains.